

Practical No-02: Text Classification using Embedding Layer and LSTM

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BBC News Topic Classification using an Embedding + LSTM Network

1. Introduction

This assignment implements a multi-class text classification system that categorizes BBC news articles into one of five topics: business, entertainment, politics, sport, and tech. The core deliverable is a deep learning model built from an Embedding layer feeding a Bidirectional LSTM network, ending in a 5-way softmax output. To contextualize its performance, the notebook also trains a classical TF-IDF + Logistic Regression baseline for comparison.

2. Dataset Information

Dataset used: BBC News dataset (Greene & Cunningham, 2006), consisting of full-length news articles published on the BBC website between 2004-2005.

Property	Value
Total articles (after cleaning)	2,127
Number of classes	5 (business, entertainment, politics, sport, tech)
business	503 (~23.7%)
entertainment	369 (~17.3%)
politics	403 (~19.0%)
sport	505 (~23.7%)
tech	347 (~16.3%)
Columns	label (category name), text (article title + body)
Class balance	Fairly balanced (~16-24% per class)
Avg. article length	~384 words (min 89, max 4,432)

The dataset is loaded from a local `bbc_news_dataset.csv` file. If the file is not present, the notebook automatically rebuilds it from the original tab-separated BBC news file hosted on GitHub, making the notebook fully self-contained and reproducible. A stratified 80/20 train-test split was used to preserve the class ratios in both sets (Train: 1,701 articles, Test: 426 articles).

3. Concepts Used

- Text preprocessing: lowercasing, URL normalization, punctuation stripping, whitespace collapsing

- Tokenization & sequence padding: Keras Tokenizer (vocabulary capped at 20,000 words, <OOV> token for rare words) and pad_sequences (fixed length of 400 tokens, chosen to cover the bulk of article lengths without excessive padding)
- Word Embeddings: a trainable Embedding layer (100-dimensional) learned from scratch
- Recurrent Neural Network: Bidirectional LSTM (64 units) capturing context from both directions of an article
- Regularization: SpatialDropout1D and Dropout layers to reduce overfitting
- Multi-class handling: integer labels via LabelEncoder, softmax output layer, sparse_categorical_crossentropy loss, and computed class_weight (balanced) used during training
- Classical ML baseline: TF-IDF vectorization (unigrams + bigrams, 20,000 features) + multinomial Logistic Regression
- Training strategy: stratified train/test split, validation split, EarlyStopping (patience = 3, restores best weights)
- Evaluation metrics: Accuracy, per-class Precision/Recall/F1, macro-averaged metrics, 5x5 Confusion Matrix, one-vs-rest ROC-AUC and ROC curves
- Exploratory Data Analysis: class distribution plots, article-length distributions, and word clouds per category
- Error analysis: inspection of misclassified articles and most-confused category pairs

4. Word Cloud Analysis

Word clouds were generated separately for each of the 5 categories using the wordcloud Python library, to visually surface the most frequent terms in each topic.

- Business — dominated by terms such as “market”, “company”, “growth”, “shares”, “profit”, “sales”, and “economy”.
- Entertainment — dominated by terms such as “film”, “best”, “actor”, “award”, “show”, “music”, and “star”.
- Politics — dominated by terms such as “government”, “labour”, “election”, “minister”, “party”, and “blair”.
- Sport — dominated by terms such as “game”, “player”, “win”, “team”, “match”, and “club”.
- Tech — dominated by terms such as “technology”, “user”, “mobile”, “net”, “digital”, and “software”.

This visual contrast confirms that vocabulary alone carries a strong, learnable signal for separating the five topics, a signal exploited by both the baseline model and the LSTM.

5. Model Architecture

Layer	Output Shape	Parameters
Embedding (vocab=20000, dim=100, len=400)	(None, 400, 100)	2,000,000
SpatialDropout1D (0.2)	(None, 400, 100)	0
Bidirectional LSTM (64 units)	(None, 128)	84,480
Dense (32, ReLU)	(None, 32)	4,128
Dropout (0.3)	(None, 32)	0
Dense (5, Softmax)	(None, 5)	165

Total trainable parameters: 2,088,773

Loss function: Sparse Categorical Crossentropy | Optimizer: Adam (learning rate = 1e-3)

Training: 9 epochs (early-stopped from a maximum of 15, restoring the best weights), batch size 32, class-weighted to balance the slightly under-represented categories (tech, entertainment).

6. Results

6.1 Baseline — TF-IDF + Logistic Regression

Category	Precision	Recall	F1-score	Support
business	1.00	0.97	0.98	101
entertainment	0.99	1.00	0.99	74
politics	0.98	0.98	0.98	81
sport	1.00	1.00	1.00	101
tech	0.97	1.00	0.99	69

Overall accuracy: 0.99 | Macro ROC-AUC (one-vs-rest): 0.9971

6.2 Deep Model — Embedding + Bidirectional LSTM

Category	Precision	Recall	F1-score	Support
business	0.9238	0.9604	0.9417	101
entertainment	0.8333	0.9459	0.8861	74
politics	0.9863	0.8889	0.9351	81
sport	0.9892	0.9109	0.9485	101
tech	0.9014	0.9275	0.9143	69

Overall accuracy: 0.9272 | Macro Precision: 0.9268 | Macro Recall: 0.9267 | Macro F1: 0.9251 | Macro ROC-AUC (one-vs-rest): 0.9903

6.3 Confusion Matrix — LSTM (test set of 426 articles)

Actual \ Predicted	business	entertainment	politics	sport	tech
business	97	1	0	0	3
entertainment	1	70	1	0	2
politics	2	4	72	1	2
sport	0	9	0	92	0
tech	5	0	0	0	64

Total misclassified: 31 out of 426 test articles (7.28%). The most common confusion pairs were sport → entertainment (9 cases) and tech → business (5 cases), suggesting some overlap in vocabulary between celebrity/lifestyle sport coverage and entertainment news, and between technology-company reporting and general business news.

6.4 Baseline vs. LSTM — Side-by-Side Comparison

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)	ROC-AUC (macro, ovr)
TF-IDF + Logistic Regression	0.99	0.99	0.99	0.99	0.9971
Embedding + Bidirectional LSTM	0.9272	0.9268	0.9267	0.9251	0.9903

Key takeaway: both models perform strongly, since the five BBC News topics use fairly distinct vocabulary. The TF-IDF baseline is competitive with — and here, ahead of — the LSTM, a reminder that deep sequence models typically need more data to clearly outperform strong classical baselines on a dataset of only ~2,100 documents. Because the model is multi-class, per-class precision/recall/F1 and the confusion matrix are more informative than a single accuracy number, since they reveal exactly which topic pairs get confused.

6.5 Custom Article Predictions (sample)

Text (truncated)	Prediction	Confidence
"The stock market rallied today after the central bank announced..."	business	0.9942
"The actress accepted the award for best performance at last night's ceremony..."	entertainment	0.9797
"The prime minister addressed parliament today, defending the government's..."	politics	0.6249
"The home team clinched the championship title after a dramatic final match..."	entertainment*	0.7022
"The company unveiled its latest smartphone, featuring a faster processor..."	business*	0.6709

**Two of the five hand-written samples were misclassified (a sport article predicted as entertainment, and a tech-product article predicted as business), both with comparatively low confidence — consistent with the sport/entertainment and tech/business confusion already visible in the confusion matrix above.*

7. Conclusion

- The BBC News dataset is a balanced, multi-class problem built from full-length news articles rather than short messages, which required adapting the pipeline: integer label encoding, a larger vocabulary and sequence length, a softmax output with sparse categorical crossentropy, a 5x5 confusion matrix, and macro-averaged one-vs-rest ROC-AUC.
- Both the classical baseline and the LSTM perform strongly, since BBC News topics have a fairly strong, learnable vocabulary signal — similar in spirit to the SMS spam finding that vocabulary alone carries a lot of the classification signal.

- Reporting the full per-class metric suite (precision, recall, F1) and the confusion matrix, rather than accuracy alone, is essential in a multi-class setting: it reveals exactly which topic pairs (sport/entertainment, tech/business) the model tends to confuse.

Possible Extensions

- Use pre-trained embeddings (GloVe/Word2Vec) instead of learning them from scratch
- Try a deeper/stacked LSTM or a GRU
- Add an attention mechanism over the LSTM outputs
- Tune MAX_LEN, VOCAB_SIZE, or use validation-set sweeps to reduce the sport/entertainment and tech/business confusion